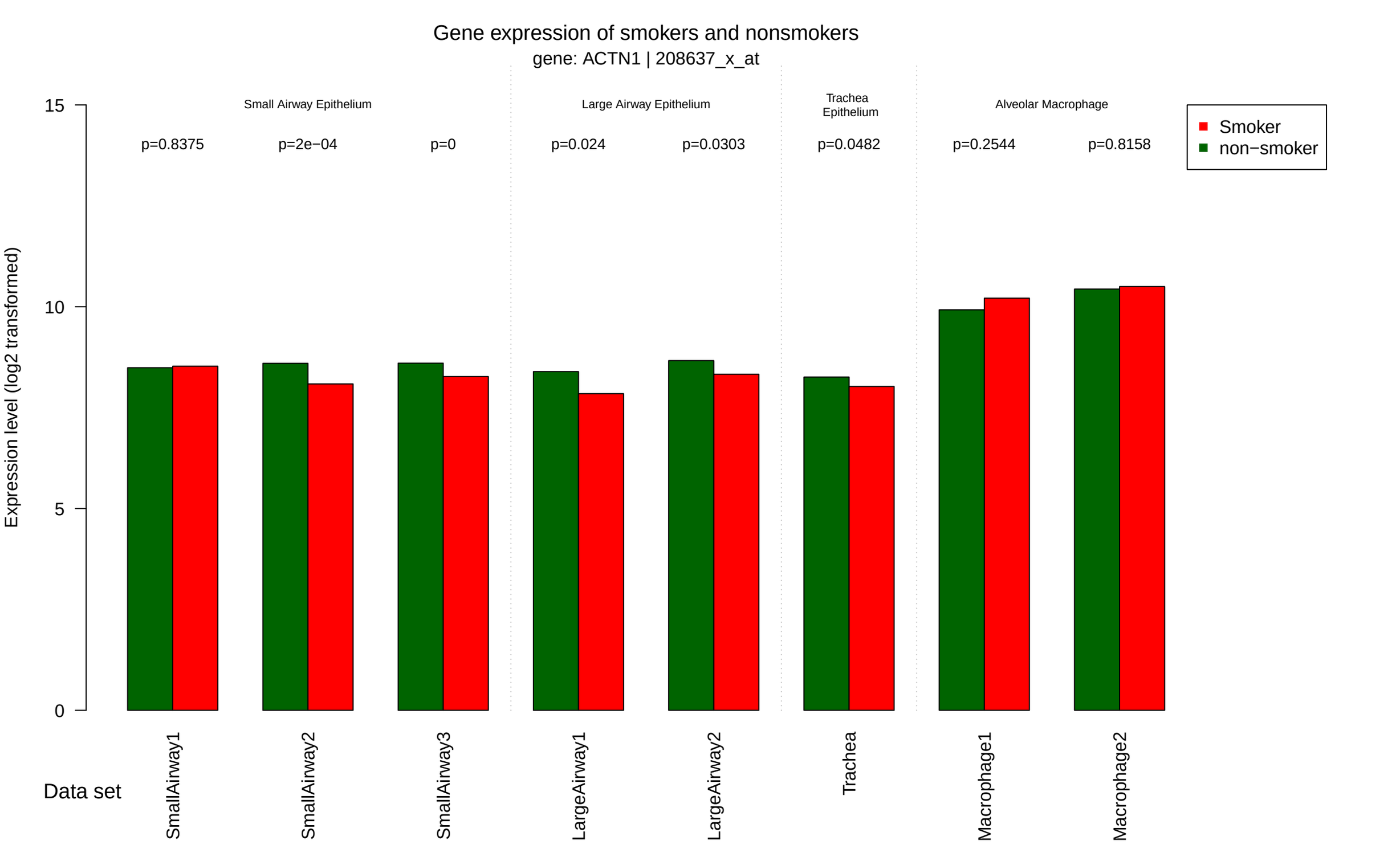
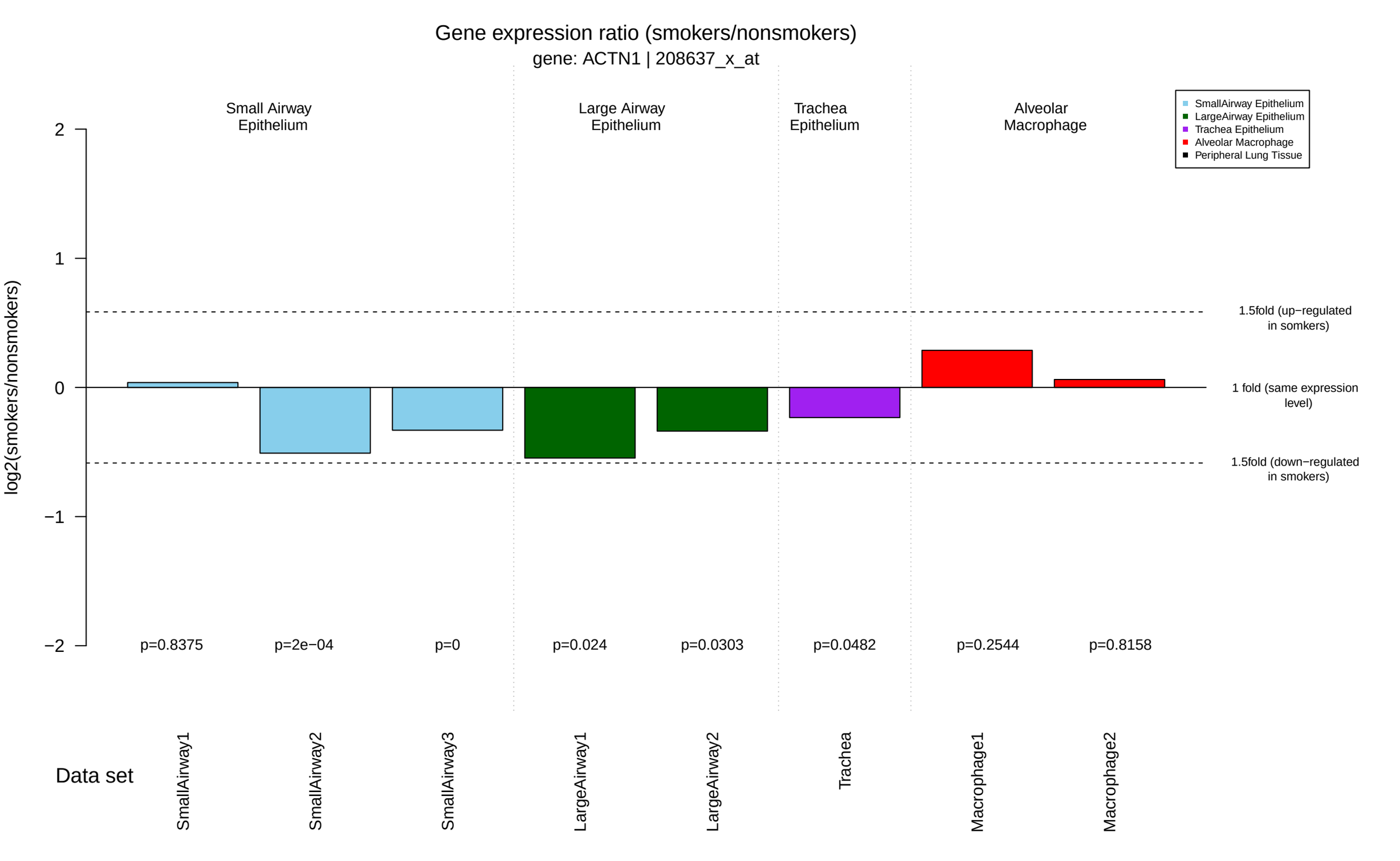
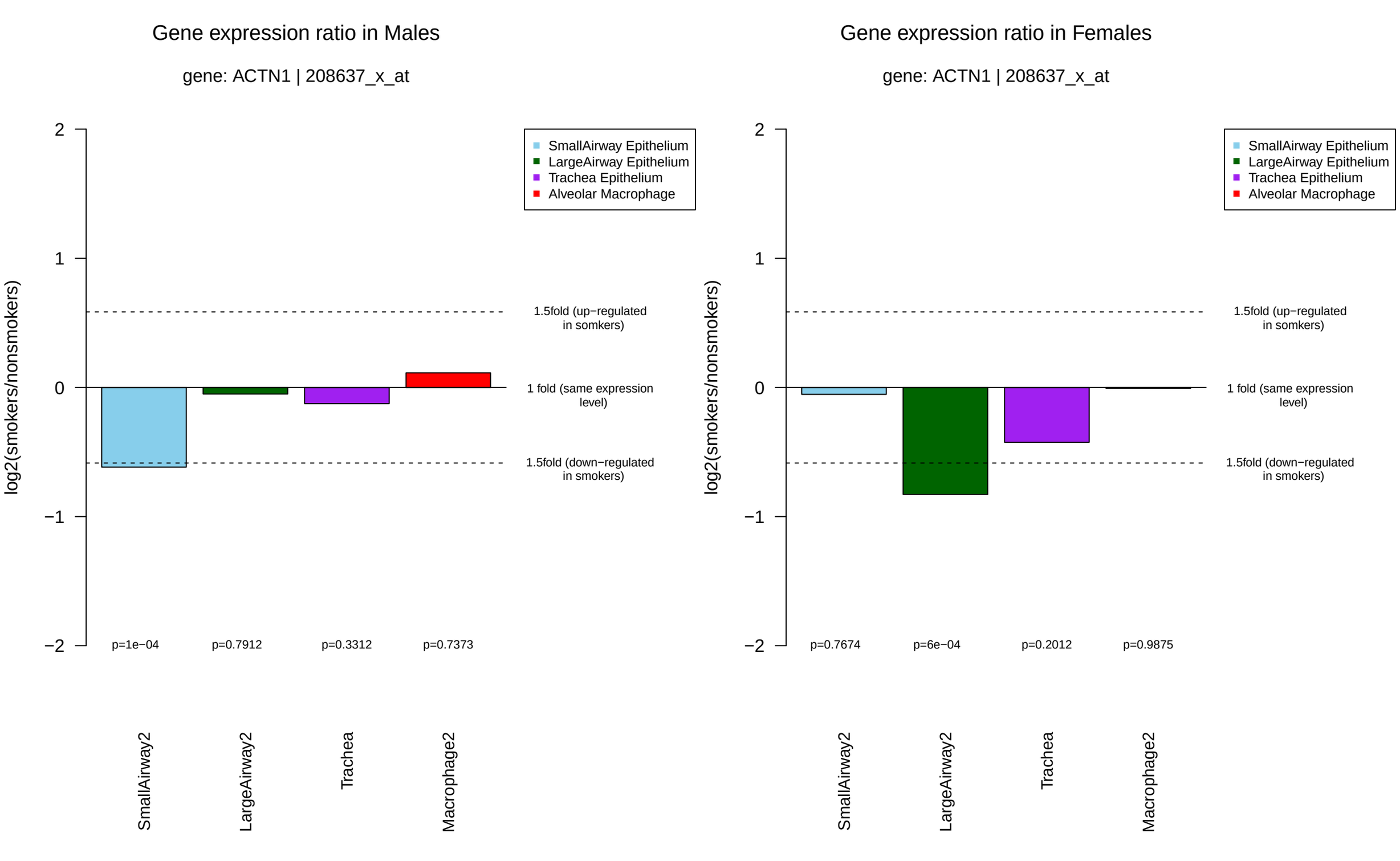
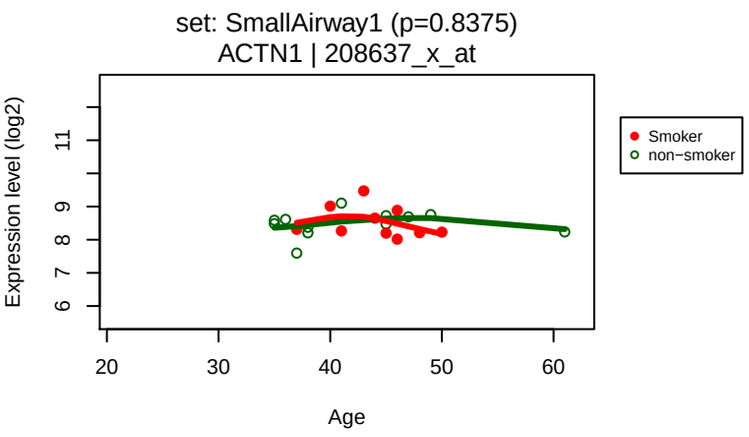
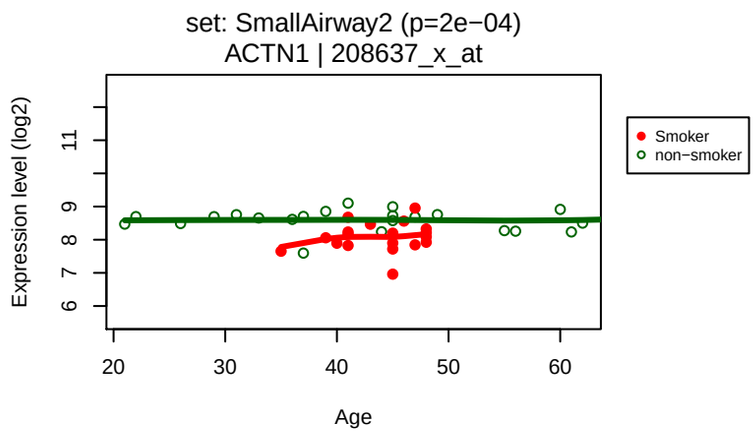
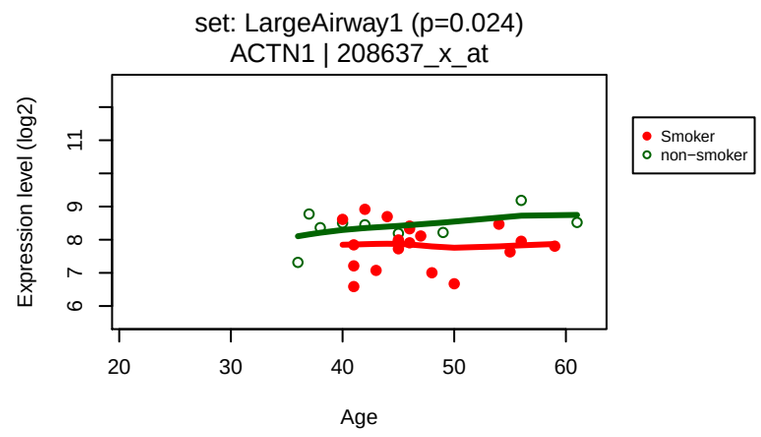

Gene expression of smokers and nonsmokers
gene: ACTN1 | 208637_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8375
p=2e−04
p=0
p=0.024
p=0.0303
p=0.0482
p=0.2544
p=0.8158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

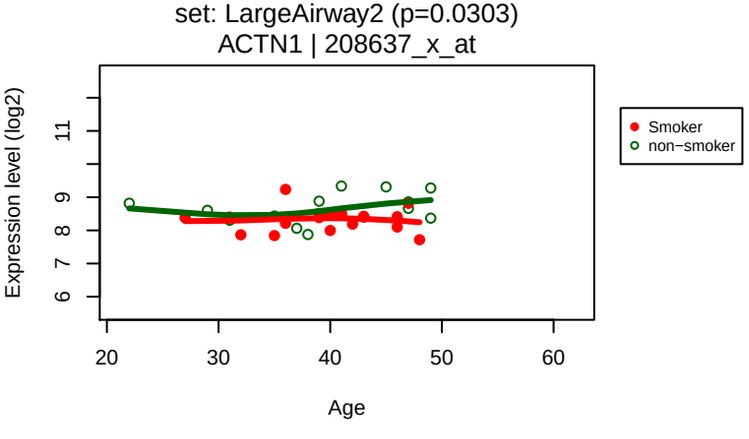

Gene expression ratio (smokers/nonsmokers)
gene: ACTN1 | 208637_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8375
p=2e−04
p=0
p=0.024
p=0.0303
p=0.0482
p=0.2544
p=0.8158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

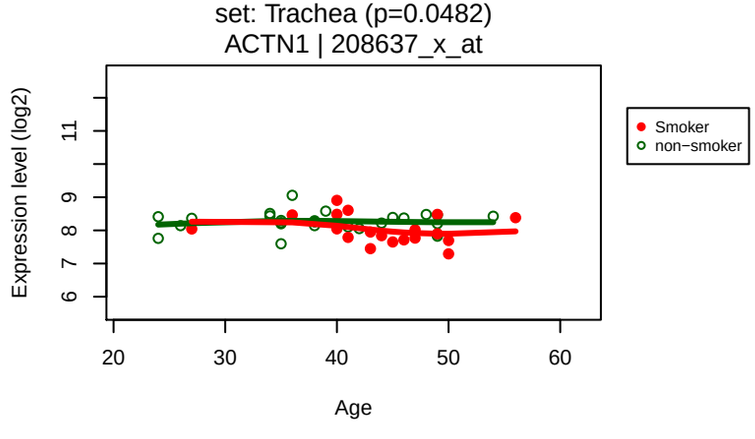

Gene expression ratio in Males
gene: ACTN1 | 208637_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.7912
p=0.3312
p=0.7373
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN1 | 208637_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7674
p=6e−04
p=0.2012
p=0.9875
SmallAirway2
LargeAirway2
Trachea
Macrophage2

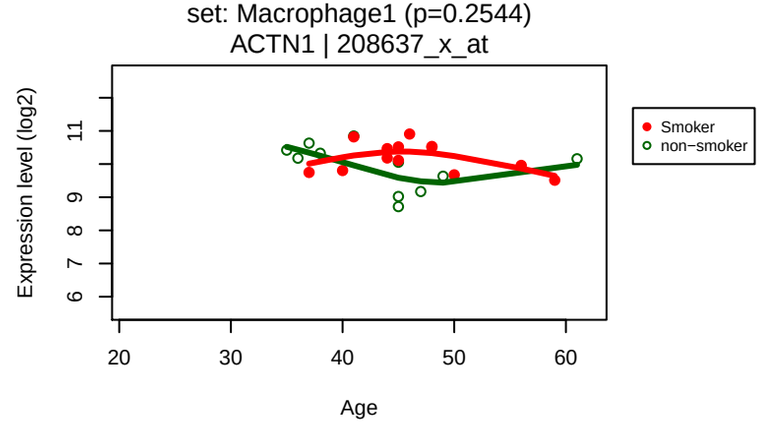

set: SmallAirway1 (p=0.8375)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker

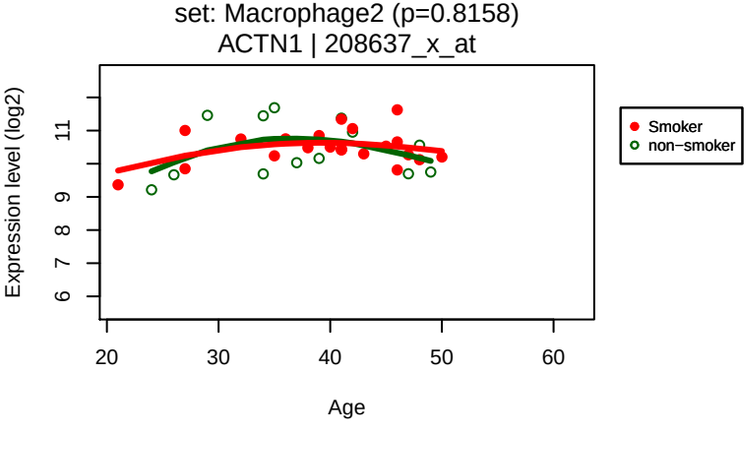

set: SmallAirway2 (p=2e−04)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker

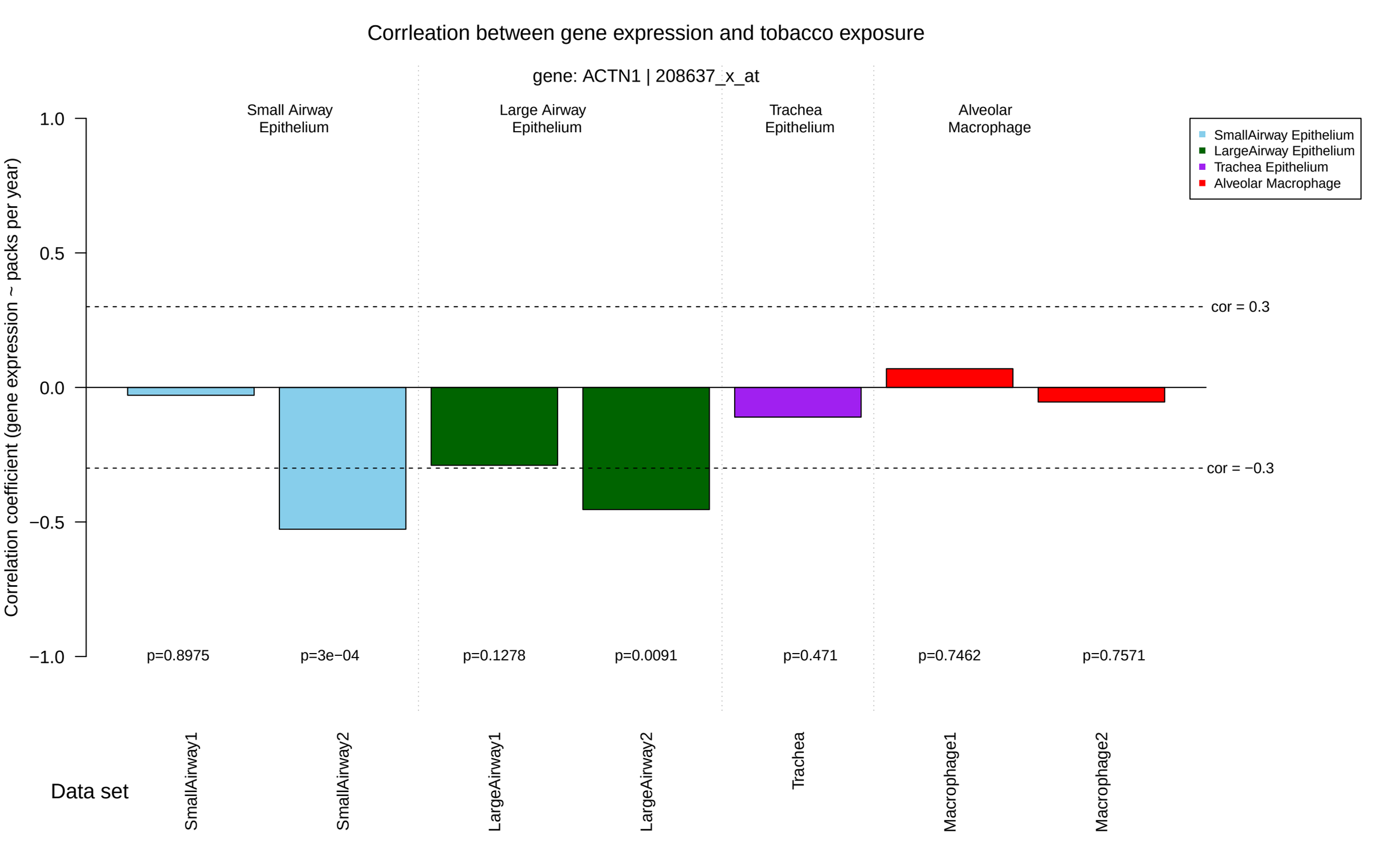

set: LargeAirway1 (p=0.024)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0303)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0482)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2544)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8158)
ACTN1 | 208637_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN1 | 208637_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8975
p=3e−04
p=0.1278
p=0.0091
p=0.471
p=0.7462
p=0.7571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2